THE TYPE OF A RATIONAL HODGE COUNTEREXAMPLE

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Abstract. The Clay Millennium statement of the Hodge conjecture is the *rational* sentence

$$\forall D\, \forall \gamma\, \exists z, \quad \mathrm{cl}(z) = \gamma,$$

with coefficients in $\mathbb{Q}$. This note records the two Lean 4 types of its negation, proves they are classically equivalent, and records that both types are uninhabited in the accompanying formalization (`BenFrohman/HodgeDisproof`). It does not produce a fourfold D_{bad} or a class γ_{bad}. The integral Hodge conjecture is a different sentence and is already false.

1. The Sentence

Let D range over smooth complex projective fourfolds and let γ range over rational Hodge classes of type $(2,2)$:

$$\gamma \in H^4(D, \mathbb{Q}) \cap H^{2,2}(D).$$

Let z range over rational algebraic cycles of codimension two and let cl be the cycle class map. The easy arrow—every such $\mathrm{cl}(z)$ is Hodge—is a theorem. The Clay problem is the converse, expressed by the universal sentence above.

Hodge's original integral sentence, with $\mathbb{Z}$ in place of $\mathbb{Q}$, is false: Atiyah–Hirzebruch (1961) produced torsion classes that are not algebraic; Kollár (1990) produced infinite-order integral classes that are not algebraic. Tensoring with $\mathbb{Q}$ kills torsion and identifies classes that differ by an integer multiple. Those terms do not inhabit the rational negation.

2. Two Types of Negation

In dependent type theory the constructive negation of a universal sentence is a function type, whereas a counterexample witness is a dependent pair (sigma/existential triple).

Lean Name	Type	Inhabitant
`RationalHodgeNegation`	$(\forall D\, \forall \gamma\, \exists z,\ \mathrm{cl}\, z = \gamma) \to \mathrm{False}$	a function
`RationalCounterexample`	$\exists D\, \exists \gamma,\ \forall z,\ \mathrm{cl}\, z = \gamma \to \mathrm{False}$	a triple $\langle D, \gamma, p\rangle$

Lean treats these as distinct types; the kernel will not accept a triple where it expects a function.

Constructive map (`counterexample_to_negation`). From a triple $\langle D_{\mathrm{bad}}, \gamma_{\mathrm{bad}}, p\rangle$ and a hypothetical proof h of the universal sentence, apply h at that host and class to obtain a cycle z, and feed z to p to obtain False. This direction requires no classical axiom.

Classical map (`negation_to_counterexample`). From a function sending every purported proof of the universal sentence to False, one does not constructively compute D_{bad}. Extracting the existential witness uses `Classical.not_forall`. The converse therefore depends on `Classical.choice`.

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The kernel-checked lemmas in `Disproof/Type.lean` are `counterexample_to_negation`, `negation_to_counterexample`, and `equivalence`. Running `#print axioms equivalence` reports `[propext, Classical.choice, Quot.sound]`. That theorem relates two types without constructing an inhabitant of either.

3. The Miss Locus

Write $L(D)$ for the rational Hodge conjecture in codimension two on a polarized fourfold D, $\Delta_{\mathrm{Hdg}}(D)$ for the primitive rational $(2,2)$ classes, and

$$\Delta_{\mathrm{miss}}(D) = \Delta_{\mathrm{Hdg}}(D) \setminus \mathrm{im}(\mathrm{cl}_D).$$

By definition,

$$L(D) \iff \Delta_{\mathrm{miss}}(D) = \emptyset.$$

On a very general hypersurface fourfold of degree $d \geq 6$ in $\mathbb{P}^5$, the Lefschetz hyperplane theorem together with monodromy gives Noether–Lefschetz vanishing

$$H^4(X, \mathbb{Q}) \cap H^{2,2}(X) = \mathbb{Q}\langle \omega^2 \rangle,$$

hence $\Delta_{\mathrm{Hdg}}(D) = 0$ and $\Delta_{\mathrm{miss}}(D) = \emptyset$. The rational Hodge conjecture holds there because no primitive rational class remains to algebraize. A large complex Hodge number $h^{2,2}$ counts complex harmonic forms of type $(2,2)$, not rational Hodge classes.

A term of `RationalCounterexample` would require a *special* fourfold with $\Delta_{\mathrm{Hdg}} \neq 0$ and $\Delta_{\mathrm{miss}} \neq \emptyset$. That host is not constructed in this note.

4. Scope: What Is Asserted and What Is Not Asserted

(1) The integral Hodge conjecture is false (classical literature counterexamples cited below).
(2) The two rational-negation types `RationalHodgeNegation` and `RationalCounterexample` are classically equivalent in Lean 4 with zero `sorry`.
(3) The constructive direction `counterexample_to_negation` uses no classical axiom.
(4) Neither rational-negation type is inhabited in this repository, nor is the universal rational sentence itself.
(5) This note does not claim the Clay Millennium Prize, and does not claim that the rational Hodge conjecture is either true or false.

References

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